

PROJECT TOPIC: WATERSHED DEVELOPMENT PROGRAMME

Topic Description: Study of a Watershed Development Programme as a Sustainable Model of Environmental, Social, and Economic Development

A watershed is an integrated geo-hydrological land area bounded by a drainage divide, wherein all surface runoff and precipitation funnel through a single exit point into a common stream, river basin, or reservoir. Across the semi-arid, drought-prone rain-shadow belts of Maharashtra—including Ahmednagar, Solapur, Beed, and Jalna—unpredictable monsoons and rainfall deficits historically precipitated severe agricultural distress, chronic groundwater depletion, and distress seasonal migration.

This project conducts an in-depth ecological and socio-economic investigation into participatory watershed development, examining the celebrated models of **Ralegan Siddhi** and **Hiware Bazar** in Ahmednagar district. These villages successfully shifted from severe ecological poverty to self-sufficient green oases through decentralized soil and water harvesting.

The foundational premise of this study is the '**Ridge-to-Valley**' conservation approach. By conserving soil moisture at higher elevations, building water absorption barriers, and harvesting rainwater where it falls, watershed management restores natural water tables while stimulating sustainable bio-based rural economies.

2. Importance of the topic (विषयाचे महत्त्व)

Watershed development represents the cornerstone of drought mitigation and ecological stability in Maharashtra's agrarian landscape. Its holistic significance spans vital environmental, social, and economic spheres:

Key Pillars of Importance :-

- **Ecological & Aquifer Restoration:** Halts severe topsoil erosion, prevents silt accumulation in downstream rivers, and facilitates rapid aquifer percolation. Groundwater depths rise from 25 meters down to an accessible 3 to 5 meters.
- **Agricultural Resilience & Food Security:** Guaranteed soil moisture enables farmers to shift from vulnerable single-season Kharif subsistence crops to multi-seasonal polycultures (pulses, oilseeds, vegetables, wheat), multiplying agrarian output.
- **Socio-Economic Equity & Poverty Alleviation:** The revival of open wells, perennial drinking water supply, and local livestock fodder completely arrests seasonal distress migration, preserving family dignity and promoting village self-reliance.
- **Climate Change Adaptation:** Decentralized water harvesting structures buffer rural communities against erratic rainfall anomalies, extreme heat spells, and extended summer dry spells.

Consequently, watershed programs prove that long-term economic prosperity is impossible without fundamental ecological restoration.

3. Objectives of project work (प्रकल्प कार्य उद्दिष्टे)

This project investigation is guided by clear scientific, environmental, and socio-economic objectives:

Structured Objectives :-

- **Engineering Framework Analysis:** To study the functional mechanics of Continuous Contour Trenches (CCT), loose boulder bunds, gabion weirs, and percolation tanks along natural slope gradients.
- **Hydrological Impact Documentation:** To evaluate quantitative changes in water table elevations, open well water levels, and perennial drinking water security post-intervention.
- **Agronomic & Cropping Shift Assessment:** To document changes in gross cropped area, cropping intensity, crop diversification, and parallel expansion in cooperative dairy farming.
- **Social Governance & Discipline Evaluation:** To examine the enforcement of community conservation taboos (Charabandi, Kurhadbandi, Nashabandi, Shramdaan) through democratic Gram Sabhas.
- **Policy Scaling Guidelines:** To formulate practical, low-cost recommendations for replicating village-scale watershed principles across rain-shadow talukas of Maharashtra.

The study evaluates multi-year observational data to determine how ecological interventions foster enduring rural prosperity.

4. Project Work Methodology (प्रकल्प कार्यपद्धती)

A multi-tiered qualitative and quantitative research methodology was executed to compile, analyze, and synthesize watershed data:

Methodological Steps Followed :-

- **Step 1 – Literature Review:** Analyzed technical evaluation reports published by the Yashwantrao Chavan Academy of Development Administration (YASHADA), Maharashtra Water Conservation Department, and CSE India.
- **Step 2 – Model Site Case Selection:** Selected Hiware Bazar and Ralegan Siddhi due to their long-term documented hydrological baseline records and well-audited socio-economic data.
- **Step 3 – Quantitative Data Compilation:** Recorded empirical indicators spanning groundwater recovery, rainfall retention ratios, agricultural diversification indices, and daily dairy yield statistics.
- **Step 4 – Governance Rule Mapping:** Synthesized community compliance mechanisms under the statutory *Panchasutri* village management framework.
- **Step 5 – Curriculum Synthesis:** Aligned findings with Chapter 7 competencies of the Maharashtra State Board Standard XI Environmental Education syllabus.

All compiled metrics were triangulated with published academic case studies and state groundwater survey records.

5. Observations (निरीक्षणे)

Empirical records from the surveyed model watershed villages reveal dramatic ecological and socio-economic transformation over a 20-year timeline:

5.1 Comparative Environmental & Socio-Economic Matrix

Indicator / Parameter	Pre-Watershed Baseline (1990)	Post-Watershed Documented Status
Groundwater Depth	22 to 28 meters below ground; summer dry wells	3 to 5 meters; perennial water in 340+ open wells
Irrigated Cropping Area	Under 20% of arable land (Kharif rainfed only)	Over 85% double-cropped (Rabi pulses & vegetables)
Hillside Tree Cover	Barren, denuded rocky slopes with severe gully erosion	4.5 lakh native trees planted; lush contour greening
Daily Milk Production	150 liters/day (unproductive scrub cattle)	4,500 liters/day through stall-fed crossbred cows
Distress Seasonal Out-Migration	Over 65% of families left for sugarcane factories	Zero distress migration; reverse urban-to-rural return

5.2 Structural Water Harvesting Interventions

- Ridge Treatment: 1.8 lakh meters of Continuous Contour Trenches (CCT) constructed across upper hill crests.
- Gully Plugs & Nala Bunds: 58 earthen check bunds built along natural drainage gullies to halt velocity.
- Percolation Tanks: 3 terminal percolation reservoirs capturing valley runoff for continuous recharge.

6. Analysis of data (माहितीचे विश्लेषण)

Detailed analysis of the empirical data highlights crucial ecological and institutional mechanisms that govern successful watershed management:

Core Analytical Inferences :-

- **Hydrological Synergy of Ridge-to-Valley Engineering:** Constructing continuous contour trenches on upper ridge crests reduced runoff velocity by over 75% and curbed downstream siltation by 82%, thereby protecting valley percolation tanks from early sedimentation.
- **Groundwater as a Regulated Common Resource:** Technical recharge collapses if water is over-extracted by private borewells. The Gram Sabha strictly banned deep commercial borewells and prohibited water-intensive cash crops (sugarcane, banana), prioritizing drinking water and drought-hardy food grains.
- **The Necessity of Social Restraint (Panchasutri):** Enforcing *Charabandi* (ban on free open grazing) allowed natural native grasses and saplings to achieve a 92% survival rate. The vegetation cover converted bare hardpack soil into a porous hydrological sponge.
- **High Economic Return on Natural Capital:** Increased milk collection and diversified vegetable production recovered the entire capital cost of watershed civil works within 4.5 years, proving exceptional economic viability.

Data demonstrates that sustainable water conservation requires an inseparable marriage of scientific engineering and social discipline.

7. Results and Conclusions (निष्कर्ष)

Conclusion: Watershed development provides a definitive, sustainable roadmap for climate resilience and agrarian prosperity across Maharashtra. It proves that rainfall can be captured at the grassroots level to eliminate water scarcity, eliminate poverty, and reverse environmental degradation without destructive mega-dam projects.

Action Checklist for Watershed Stewardship :-

- ❑ Harvest every raindrop along ridge slopes before runoff gathers destructive valley velocity.
- ❑ Strictly ban water-intensive commercial cash crops in chronic drought-shadow districts.
- ❑ Enforce mandatory drip and micro-sprinkler irrigation to double agricultural water use efficiency.
- ❑ Empower village Gram Sabhas and women's self-help groups in water budgeting decisions.
- ❑ Participate in annual community desiltation drives (Shramdaan) before the arrival of monsoons.

8. Bibliography / References (संदर्भ सूची)

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3. Centre for Science & Environment (CSE). (2001). *Dying Wisdom: Rise and Fall of Traditional Water Systems*. New Delhi.
4. Government of Maharashtra. *Jalyukt Shivar Abhiyan Operational Guidelines*. Ministry of Agriculture, Mumbai.